

A_∞ Enhancement of Geometry

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The goal of our research is to prove homological mirror symmetry conjecture or to develop the relevant area of study. The claim of homological mirror symmetry conjecture is A_∞ -equivalence of Fukaya category $Fuk(X)$ and A_∞ -enhancement $D^b(Coh(X))$ of derived category of coherent sheaves. This statement says the elaboration of mirror symmetry, originally correspondence of symplectic data on Hodge structure and that of complex in Calabi-Yau manifold, and we named symplectic side as A-model, the complex side as B-model, mirror symmetry solves many of QFT problem as well as Landau-Ginzburg model and others, and homological mirror symmetry is generalization of mirror symmetry in ∞ -category, but the proof of homological mirror symmetry isn't easy.

One motivation to study A_∞ -algebra is to study Fukaya category.

A_∞ algebra is a generalization of commutative ring, and I'll use A_∞ algebra to describe algebraic geometry. Recall a commutative ring defines an affine scheme by its ring spectrum $X = V(\mathfrak{p})$ with its structure sheaf $\mathcal{O}_X$, and that can also be described functorially as $F : CRing \rightarrow Set$. Hereafter I focus on functorial point of view.

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The proof of homological mirror symmetry conjecture is discussed from multi-dimensional perspective, including counting Gromow-Witten invariant, or Balmer spectrum, while generalizing formal geometry of non-unital A_∞ -algebra to A_∞ -category.

1 Geometric Realization of A_∞

Definition 1.1. Counital Coalgebra

Coalgebra B is a graded vector space with morphism $\Delta : B \rightarrow B \otimes B$ s.t. $(\Delta \otimes id) \circ \Delta = (id \otimes \Delta) \circ \Delta$ called coproduct, and counital coalgebra is coalgebra B with unit $\epsilon : B \rightarrow 1$ where 1 is the unit object of $\mathcal{C}$, satisfying $(\epsilon \otimes id) \circ \Delta = (id \otimes \epsilon) \circ \Delta$. Counital coalgebra B is cofree counital coalgebra if it's given by $B = C(V) \cong T^c(V)$.

Definition 1.2. A Non-Commutative Thin Scheme

A non-commutative thin scheme X is a functor $X : Alg_{\mathcal{C}^f} \rightarrow Sets$, that is isomorphic to $X \cong Hom(-^*, B_X)$ for some coalgebra B_X , where $\mathcal{C}^f$ is artinian category of finite dimensional objects in $\mathcal{C}$, where $\mathcal{C}$ is $\mathbb{Z}$ -graded vector spaces. In particular, X is a non-commutative formal manifold, if the functor X is isomorphic to some $Spc(T(V))$ for a cofree counital tensor coalgebra $T(V) = \bigoplus_{n \geq 1} V^{\otimes n}$. The dimension of X is defined as $dim_k V$.

Definition 1.3. A Non-Commutative Thin DG-Scheme

A non-commutative thin dg-scheme is a pair (X, d) where X is a non-commutative thin scheme, and d is a vector field on X of degree $+1$ satisfying bracket of dgla as $[d, d] = 0$. This d is called homological vector field.

Definition 1.4. Non-Commutative Formal Pointed DG-Manifold

A non-commutative formal pointed dg-manifold is a pair $((X, x_0), d_X)$ such that (X, x_0) is a non-commutative formal pointed graded manifold, and d_X is a homological vector field on X such that $d_X|_{x_0} = 0$.

Proposition 1.1. Taylor Decomposition of Graded Vector Field

For the case of formal manifold X , let the corresponding cofree counital coalgebra be $B_X \cong T^c(V)$, and the homological vector field be coderivation $d_X : B_X \rightarrow B_X$, that can have restriction by cofreeness, thus by using projection, $\pi \circ d_X : B_X \rightarrow V[1]$. If in addition, d_X can be decomposed degreewise for its linearity, as $d_X = \sum_{n \geq 1} d_X^{(n)}$ for linear maps $d_X^{(n)} : V^{\otimes n} \rightarrow V[1]$.

Definition 1.5. Non-Unital A_∞ Algebra

A non-unital A_∞ -algebra $A = (A, m_n)$ is a graded vector space A and coherence condition $m_n : A^{\otimes n} \rightarrow A[2 - n]$ with Stasheff relation, whose geometric interpretation is the structure of a nc fp dg-manifold $((X, x_0), d_X)$ together with a coalgebra isomorphism $B_X \cong T(T_{x_0}X)$ with $x_0 : k \rightarrow B_X$.

Definition 1.6. Structure of A_∞ Algebra

The structure of A_∞ Algebra on $V \in Ob(Vect_k^{\mathbb{Z}})$ is given by coderivation d of degree $+1$ of the non-counital cofree coalgebra $T_+(V[1]) = \bigoplus_{n \geq 1} V^{\otimes n}$ s.t. $[d, d] = 0$.

Proposition 1.2. Stasheff Relation

The condition of nc fp dg-manifold $((X, x_0), d_X)$ derives Stasheff relation of the A_∞ algebra, that is $m_n := d_X^{(n)}$ for $m_n : A^{\otimes n} \rightarrow A$ s.t. $\sum_{r+s+t=n} (-1)^{r+st} m_{r+1+t}(id^{\otimes r} \otimes m_s \otimes id^{\otimes t}) = 0$, whose degree is $|m_n| = 2 - n$. The Taylor decomposition of homological vector field $\pi \circ d_X = \sum_{n \geq 1} d_X^{(n)}$ is degreewise by linearity, as $d_X^{(n)} : B_X^{(n)} \rightarrow B_X^{(1)}$.

Definition 1.7. Non-Commutative Formal n-Pointed DG-Manifold

Nc formal n -pointed dg-manifold $(\mathcal{X}, d_{\mathcal{X}}, \{pt_i\})$ is nc formal graded dg-manifold $(\mathcal{X}, d_{\mathcal{X}})$, and the points are $pt_i \in \mathcal{X}(k)$ for $1 \leq i \leq n$, and coderivation satisfies $d_{\mathcal{X}}|_{pt_i} = 0$, $d_{\mathcal{X}}|_{TX_i} : TX_i \rightarrow \bigoplus_{j,k} TX_j \oplus TX_k$ where X_i is formal neighborhood of X at pt_i .

Definition 1.8. A_∞ -Category

A_∞ -category $\mathcal{A}$ is nc formal dg-manifold $(\mathcal{X}, d_{\mathcal{X}}, \{pt_i\})$ with its coalgebra $B_{\mathcal{X}} \cong T^c(V[1])$. This V has a decomposition $V = \bigoplus_{i,j} E_{ij}$, where $E_{ij} := T(\widehat{X_i \times X_{j_{pt_i, pt_j}}})$.

Remark 1.1. T

is V is a k -linear associative algebra generated from k -linear category $\mathcal{C}$: let $Ob(\mathcal{C}) = I$ to define an algebra $V = \bigoplus Hom(i, j)$ for $i, j \in I$ is a k -linear algebra with the unit $1_V = \sum_{i \in I} id_i$, where $\pi_i = id_i \in V$ is a commuting idempotent i.e. $\pi_i^2 = \pi_i$, $\pi_i \pi_j = \pi_j \pi_i$. This makes homomorphism $k^I \rightarrow V$ of algebra with unit.

Proposition 1.3. Stasheff Relation

From non-unital A_∞ category $\mathcal{A}$, the coalgebra $d_\mathcal{X} : T^c(\oplus E_{ij}[1]) \rightarrow T^c(\oplus E_{ij}[1])$ satisfies the Stasheff relation. Consider the tensor coalgebra can be given by the grading structure $T^c(\oplus_{1 \leq i, j \leq n} E_{ij}[1]) = \oplus_{m \geq 0} \oplus_{1 \leq i_0, i_1, i_2, \dots, i_m \leq n} E_{i_0 i_1} \otimes E_{i_1 i_2} \otimes \dots \otimes E_{i_{m-1} i_m} := \oplus_{m \geq 0} B_\mathcal{X}^m$. From cofreeness of the coalgebra, the coderivation $d_\mathcal{X}$ canonically projected onto $\pi : B_\mathcal{X} \rightarrow B_\mathcal{X}^1$, and that can be decomposed linearwise, so $\pi \circ d_\mathcal{X} = \Sigma d_\mathcal{X}^n$ where $d_\mathcal{X}^n : B_\mathcal{X}^n \rightarrow B_\mathcal{X}^1$, and we have Stasheff relation from $[d_\mathcal{X}, d_\mathcal{X}] = 0$, and $m_n := d_\mathcal{X}^n$.

Remark 1.2

$d_\mathcal{X} \circ \eta = 0$ implies $d_\mathcal{X} \circ \eta(1) = 0$ or $d_\mathcal{X}(1) = 0$. This means curvature is $m_0 = 0$.

Remark 1.3

For the case of $I = \{i\}$ consists of one vertex, we write $d|_{pt} = d \circ \eta$.

For the case of infinite objects A_∞ category is its inductive limit of finite objects.

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